

# Food Demand Forecasting

*Teodora Atanasova & Martina Puglisi*

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# RESEARCH QUESTION

**"Does clustering fulfillment centers  
by operational profile reduce demand  
prediction error and procurement waste?"**

# RESEARCH QUESTION

## CORE HYPOTHESIS

centers sharing similar operational profiles also share similar demand patterns  
=> model trained on separate centers will learn those localized patterns more accurately

## CORE UNCERTAINTY

Sufficiently powerful models could be capable of learning group-specific patterns implicitly without explicit pre-segmentation

## COMPARISON

### LVL A) GLOBAL APPROACH

- **FRAMEWORK:** Single model trained on the entire unified dataset (456,548 rows).
- **INFORMED POOLING:** Integrates cluster\_id as an explicit categorical feature instead of training blindly.
- **CORE HYPOTHESIS:** Maximizes statistical power and global trends, allowing the trees to isolate group-specific patterns via feature splits without losing data volume.

### LVL A) SEGMENTED APPROACH

- **FRAMEWORK:** Dataset physically partitioned into 3 distinct, independent sub-matrices.
- **NO POOLING:** Separate models trained in complete isolation for each operational cluster.
- **CORE HYPOTHESIS:** Eliminates cross-group variance and noise, forcing the model's internal capacity to specialize 100% on localized demand dynamics.

# COMPLETE PIPELINE

## PHASE A: Data preparation and clustering

**01**

Data Loading &amp; Initial EDA

**02**

Cleaning and preprocessing

**03**

Multi-variate explorative analysis

**04**

Feature engineering pipeline

**05**

Unsupervised clustering

**06**

Stratified subsampling

## PHASE B: Predictive ML Modeling

**01**

Metric Setup (RMSLE) &amp; Target Splitting

**02**

Naive Temporal Baselines

**03**

Parametric Linear Baselines

**04**

LVL A: Global Non-Linear Models (HGBR)

**05**

LVL B: Cluster-Segmented Modeling

**06**

Robustness Probes and Business Cost

# DATASETS DESCRIPTION

train.csv - 456,548 rows

| Column Name           | Type   | Range / Values    | Business/ML Role                         |
|-----------------------|--------|-------------------|------------------------------------------|
| id                    | int    | 1.0 MLN → 1.5 MLN | Unique identifier (dropped after load)   |
| week                  | int    | 1 → 145           | Chronological week timeline (~2.8 years) |
| center_id             | int    | 10 → 186          | Join key to fulfillment metadata         |
| meal_id               | int    | 1,062 → 2,956     | Join key to meal metadata                |
| checkout_price        | float  | 2.97 → 866.27     | Final price paid; used as Revenue Proxy  |
| base_price            | float  | 55.35 → 866.27    | Undiscounted standard price; Cost Proxy  |
| emailer_for_promotion | binary | 0 or 1            | Sparse indicator for email campaigns     |
| homepage_featured     | binary | 0 or 1            | High-visibility app/web highlight banner |
| num_orders            | int    | 13 → 24,299       | Target Variable (highly right-skewed)    |

fulfilment\_center\_info.csv - 77 rows

| Column Name | Type  | Range / Values | Business/ML Role                           |
|-------------|-------|----------------|--------------------------------------------|
| center_id   | int   | 10 → 186       | Primary Key for spatial operations         |
| city_code   | int   | 456 → 713      | Anonymized geographic city marker          |
| region_code | int   | 23 → 93        | Anonymized regional cluster identifier     |
| center_type | cat   | TYPE_A, B, C   | Categorical classification of facility ops |
| op_area     | float | 0.90 → 7.00    | Physical center footprint size metrics     |

meal\_info.csv - 51 rows

| Column Name | Type | Unique Values     | Business/ML Role                                  |
|-------------|------|-------------------|---------------------------------------------------|
| meal_id     | int  | 51 unique IDs     | Primary Key for catalog properties                |
| category    | cat  | 14 distinct types | Dish style (Beverages, Pizza, Pasta, etc.)        |
| cuisine     | cat  | 4 traditions      | Thai, Indian, Italian (most ordered), Continental |

*the 3 datasets have been merged together into one, using as keys meal\_id and center\_id*

# DATASETS DESCRIPTION

## ANONYMIZATION

all identifiers (center\_id, meal\_id, city\_code, region\_code, center\_type) are anonymous

## NO TIMESTAMPS

only relative week numbers are available, can't anchor to real calendar events

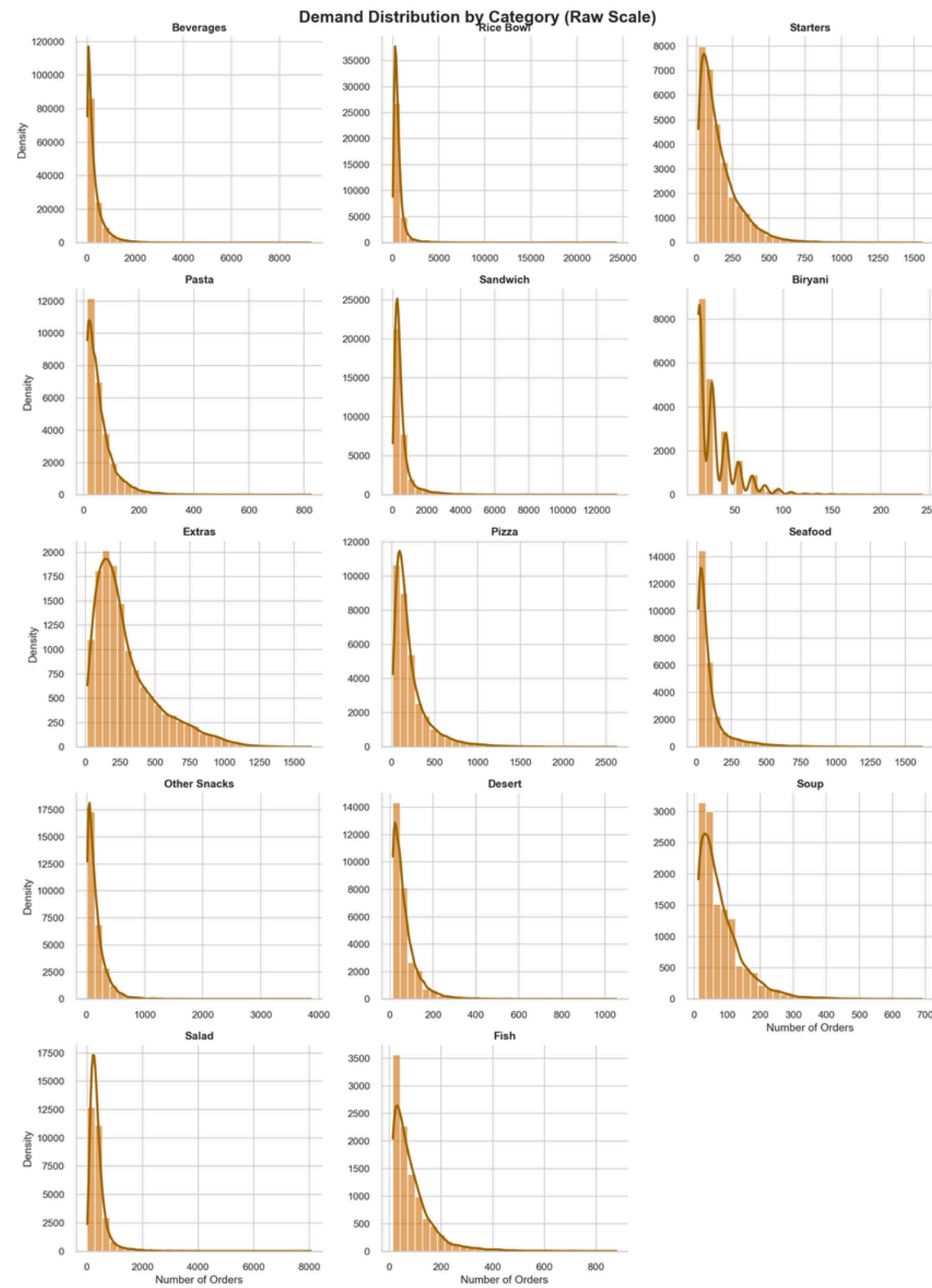
## NO EXPLICIT COSTS

"base\_price" and checkout\_price are used as proxies in the business impact analysis

## NO ZERO-ORDER ROWS

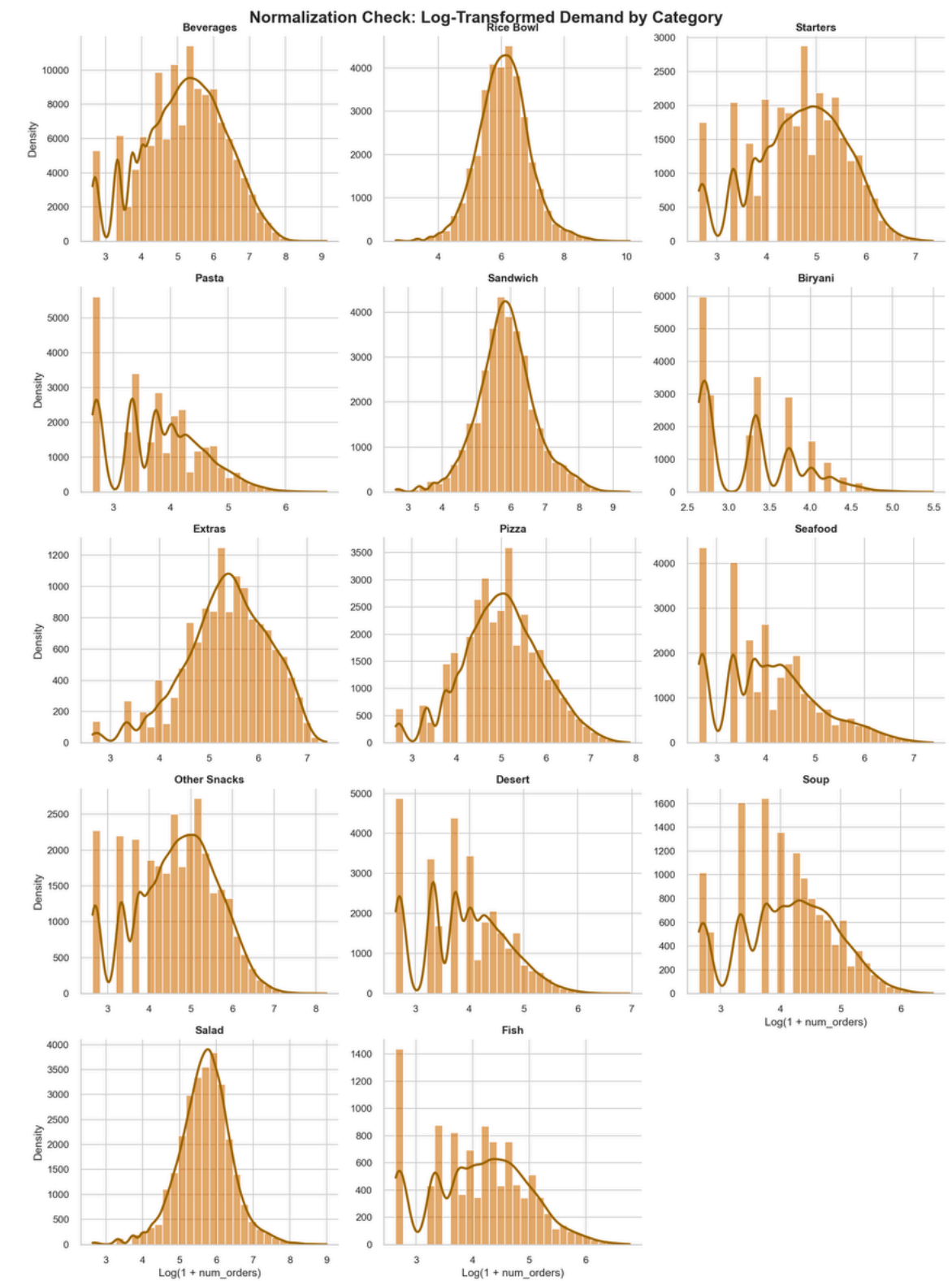
weeks with 0 orders for a center-meal are absent from the training data, rather than recorded as 0

# DATA TRANSFORMATIONS



## LOGARITHMIC TRANSFORMATION

of the target variable  
“num\_orders”



# FEATURE ENGINEERING

Converting raw relational schemas into deliberate mathematical representations of market dynamics.  
Added into the main training dataset

## CROSS SECTIONAL FEATURES

### **discount\_amount**

Absolute raw price reduction. Strongest baseline drivers

### **discount\_pct**

Absolute raw price reduction. Strongest baseline drivers

### **discount\_flag**

Absolute raw price reduction. Strongest baseline drivers

### **combined\_promo**

Absolute raw price reduction. Strongest baseline drivers

## TEMPORAL FEATURES

### **lag\_4, lag\_8**

medium-term demand cycles per center-meal pair

### **rolling\_mean\_12**

Long-window macro trajectory tracker

### **rolling\_std\_4**

Short-windows local transaction volatility & noise index

### **week\_of\_year**

Cuclical calendar index capturing annual seasonality



**Leakage Prevention** via **Temporal Offsets**  
to prevent lookahead bias, by doing  
.shift(1)

# CLUSTERING

## SCALE PREDICTABILITY

- **The Metric:** Combines **average weekly volume** (mean\_demand) with its **stability over time**.
- **The Safeguard:** Uses the **Coefficient of Variation** instead of raw volatility.
- **The Goal:** **Neutralizes the "size effect"**—preventing massive regional hubs from dominating the geometric distance and drowning out small fulfillment centers.

## PROMO SENSITIVITY

- **The Metric:** Calculates a **demand multiplier** showing **how much business surges when a marketing campaign hits**.
- **The Data Challenge:** Quiet centers with zero historical promotion exposure create missing data gaps (NaN).
- **The Fix:** **Missing metrics are neutralized to 1.0** (the baseline for "no change"), preventing the algorithm from misclassifying untested sites as failures.

## PROMO SENSITIVITY

- **The Metric:** Uses the **correlation between price and demand** to track how **sensitive** customers are to cost changes.
- **The Logic:** A **strong negative correlation flags price-sensitive markets**; a flat correlation flags inelastic areas where customers buy regardless of price.
- **The Goal:** **Separates regions where discounts drive massive volume** from strict, inelastic constraints where price changes have zero impact.

## PHYSICAL CONSTRAINTS

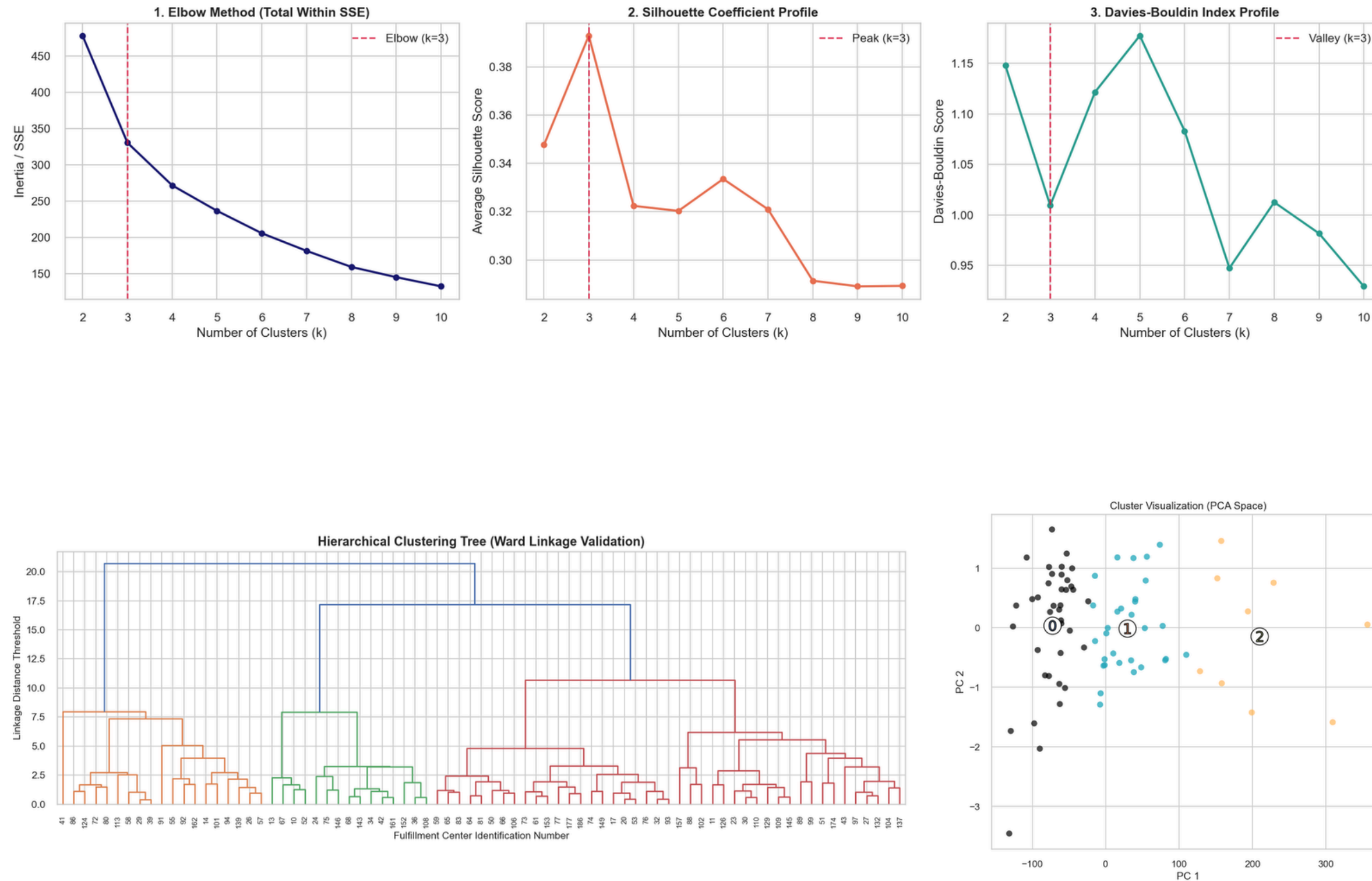
- **The Metric:** Maps the concrete reality of each facility using its **physical size (op\_area)** and **facility type**.
- **The Logic:** customer behaviors (like buying spikes) are limited by physical infrastructure. A small urban outpost cannot process the volume of a massive regional warehouse.
- **The Goal:** **ensuring clusters respect the physical capacity and structural limits** of each site.

## FINAL UNIFIED FEATURE ARRAY MATRIX:

- mean\_demand
- hoemapge\_uplift
- price\_elasticity\_proxy
- center\_type
- coefficient\_of\_variant
- emailer\_uplift
- op\_area
- city\_region\_markers

# CLUSTERING

## Unsupervised Internal Validation Indices for Optimal k Selection



- > **SSE:** within-clster compactness
- > **Silhouette-score:** measures cluster. assignment confidence. Higher is better
- > **Davies-Bouldin Index:** ratio of within-cluster distances to between-cluster distances. Lower is better

**k = 3**

# BUSINESS INTERPRETATION

| Cluster ID | Mean Weekly Demand | Volatility (CV) | Email Sensitivity | Homepage Sensitivity | Price Elasticity | Operational Area | Dominant Center Type | Network Count |
|------------|--------------------|-----------------|-------------------|----------------------|------------------|------------------|----------------------|---------------|
| Cluster 0  | 180.492            | 1.332           | 2.720             | 2.827                | -278             | 3.400            | TYPE_A               | 38 Centers    |
| Cluster 1  | 282.605            | 1.403           | 3.084             | 2.689                | -313             | 4.277            | TYPE_A               | 30 Centers    |
| Cluster 2  | 462.479            | 1.302           | 2.846             | 2.545                | -377             | 5.489            | TYPE_B               | 9 Centers     |

## CLUSTER 0 n= 38 STANDARD BASELINE

- Small-to-moderate footprint facilities
- lower average demand.
- consistent baseline volatility
- predictable, standard marketing reactivity.

## CLUSTER 1 n= 30 PROMO-SENSITIVE

- Mid-sized hubs
- highest overall demand volatility.
- intensely reactive to marketing initiatives, with an exceptional 3.08x demand surge when targeted by email promotions.

## CLUSTER 2 n= 9 HIGH-VOLUME HUBS

- Massive regional fulfillment complexes
- highest volume operations.
- low, stable volatility margins
- strong macroeconomic exposure via elevated price elasticity constraints (-0.377).

# SUBSAMPLING

GOAL: reduce the number of rows in the dataset to speed up training

## STRATIFIED SUBSAMPLING

- **cluster id**

stratification key

- **cuisine type**



prevents data scarcity for some categories



controls that cuisine buckets maintain critical mass variance and representation

**rows = 300 000**

used for main pipeline

## 2 SUBSAMPLES

- **real proportions**

based on cluster\_id

- **forced balanced**



preserves the real data-generating process



used as a robustness check against cluster-size imbalance

# Predictive Modelling

## PHASE B — PREDICTIVE MODELLING

“Does segmenting fulfillment centers by operational profile reduce demand prediction error and procurement waste?”

### LEVEL A — GLOBAL

One model on the full subsample;  
cluster\_id given as a feature.  
Maximises statistical power.

### LEVEL B — SEGMENTED

Data physically split by cluster;  
one HGBR trained in isolation per  
operational archetype.

### LINEAR NULL

Linear & Ridge as the empirical  
floor. Every model is judged on  
chronological-holdout RMSLE.

# HOW WE MEASURE & VALIDATE

1

## STATISTICAL METRIC

$$\text{RMSLE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\log(1 + \hat{y}_i) - \log(1 + y_i))^2}$$

Penalises proportional error.  
A 20% miss costs the same on low- and high-volume meals, mirroring procurement waste.

2

## CHRONOLOGICAL SPLIT

**Train = weeks 1–135**  
**Validation = weeks 136–145**

Strictly forward-looking.  
A random split would leak future weeks into training and inflate accuracy.

3

## OPERATIONAL METRICS

$$\text{Waste} = \sum_{i=1}^n \max(0, \hat{y}_i - y_i) \cdot \text{checkout\_price}_i$$

$$\text{Lost Sales} = \sum_{i=1}^n \max(0, y_i - \hat{y}_i) \cdot \text{base\_price}_i$$

Reveals the asymmetry RMSLE hides:  
all good models are shortage-biased,  
i.e. lose more revenue from stockouts  
than they waste in excess

# NAÏVE BASELINES: THE NULL FLOOR

Two training-free rules set the floor every learned model must beat

LAST-WEEK · predict = log(orders  $t-1$ )

**0.7072**

A single recent week is a noisy predictor.

ROLLING MEAN · 12-week average

**0.6284**

≈ 11% better — demand follows its trend.

## THE BENCHMARK TO BEAT: 0.6284

Any model — linear, ensemble, or segmented — must improve on the rolling-mean rule to justify its complexity over something a procurement analyst could compute by hand. The gap between the two rules already shows demand at the center-meal level is driven by medium-term behaviour, not week-to-week noise.

# LINEAR & RIDGE REGRESSION

The additive null hypothesis — does modelling price & promotion beat the floor?

Linear Regression

0.5270

Ridge (CV-tuned  $\alpha = 1000$ )

0.5271

≈ 16% GAIN OVER THE NAÏVE FLOOR

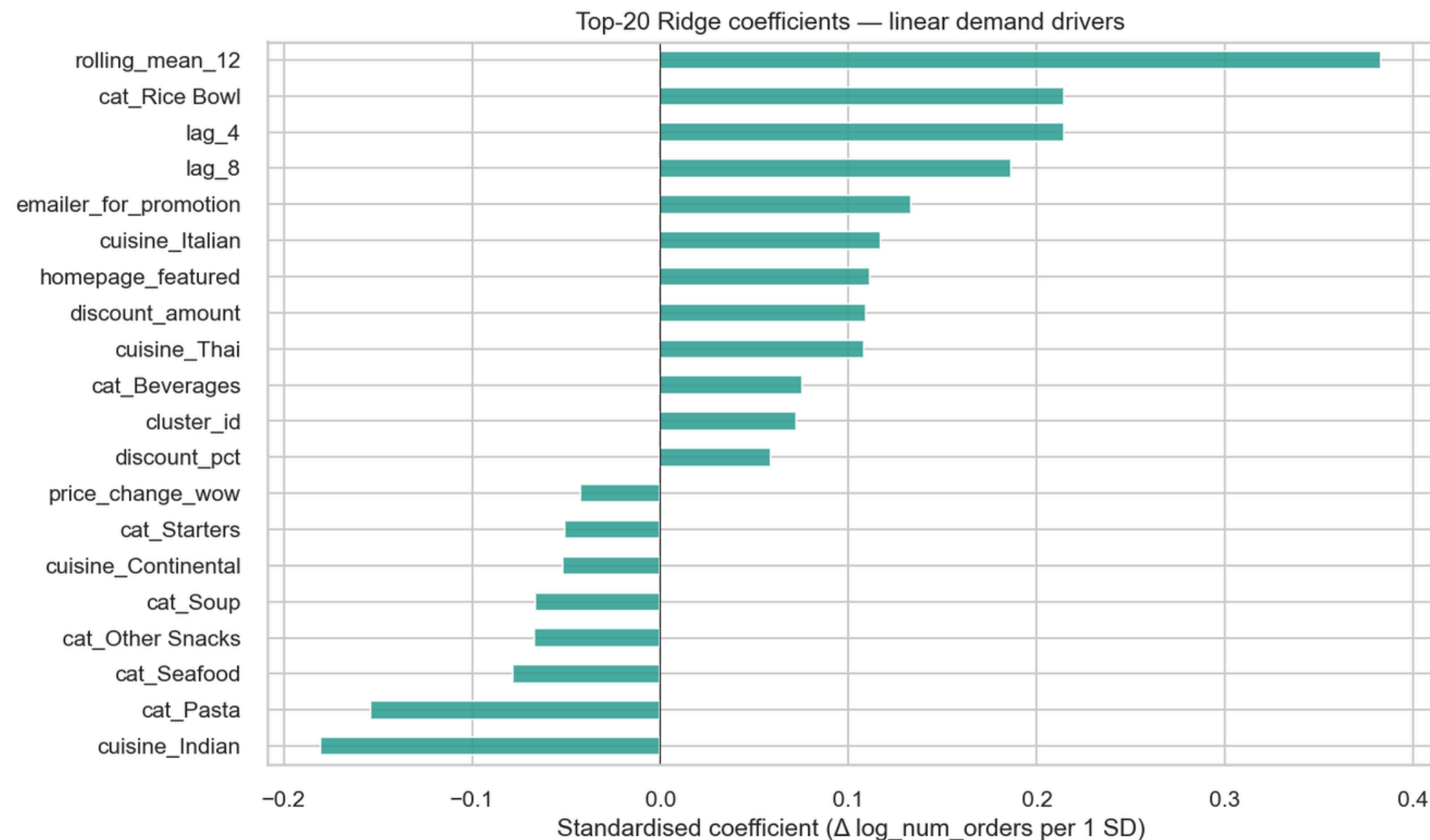
Modelling price & promotion adds real, measurable signal.

## WHAT RIDGE TELLS US

- **Strong regularisation ( $\alpha = 1000$ )** selected at the top of the grid — a signal of heavy collinearity among price & temporal features.
- **Ridge  $\approx$  Linear ( $-0.0002$ )** means regularisation buys coefficient stability, not accuracy. We read drivers off Ridge for that reason.
- **Additive structure is the ceiling** and each feature acts independently — real demand has interactions linear models cannot represent.
- **Motivates non-linear ensembles** to capture price  $\times$  promotion  $\times$  temporal interactions the linear model misses.

# TOP-20 RIDGE COEFFICIENTS

Standardised coefficients — the strongest linear demand drivers



## READING THE WEIGHTS

- **rolling\_mean\_12 dominates ( $\beta \approx 0.40$ )** - recent demand history carries most of the linear signal — validating the temporal feature engineering.
- **lag\_4 & lag\_8 follow** - medium-term demand cycles are the next strongest predictors.
- **Menu effects appear next** - cat\_Rice Bowl, cuisine\_Italian/Indian shift demand by category.
- **city\_code, discount\_flag  $\approx 0$**  - under strong shrinkage Ridge ignores predictors with no marginal linear value.

# NON-LINEAR ENSEMBLES (LEVEL A)

Random Forest vs HGBR — two architectures across the bias–variance trade-off

## RANDOM FOREST

Parallel bagged trees aggregated by averaging — primarily variance reduction.

**0.5038** RMSLE

## HGBR

Sequential trees correcting residuals — primarily bias reduction. Histogram bins keep it fast.

**0.4984** RMSLE

## THEY CONVERGE

Only 0.006 RMSLE apart — inside the noise floor. Two opposite mechanisms reach the same accuracy: the pipeline is nearing the dataset's **predictability ceiling**.

## TUNING (RF, TimeSeriesSplit CV)

10 sampled configs · carried forward as Level A

CV RMSLE range

**0.519–0.547**

Best config

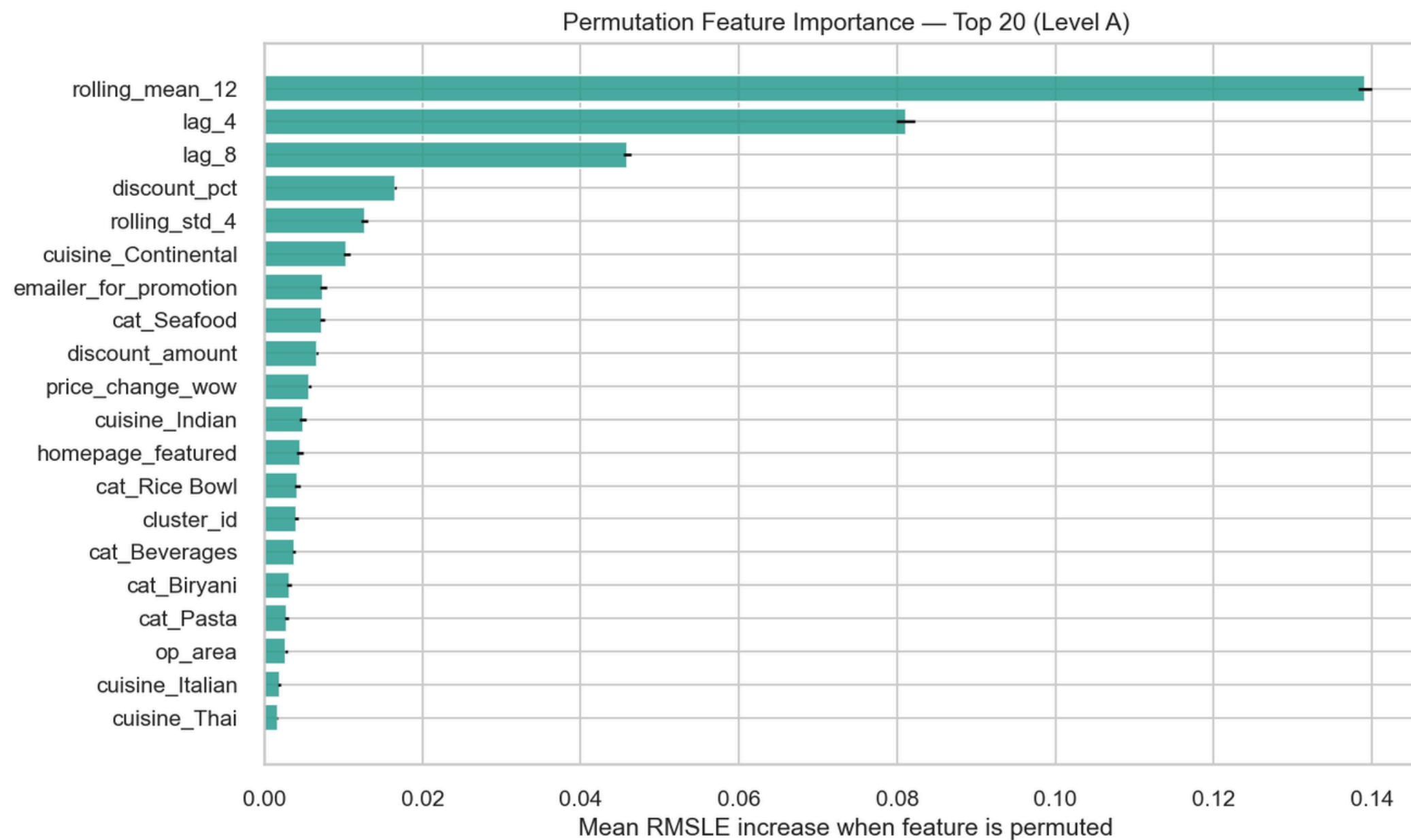
**0.5190 ± 0.011**

Tuning gain

**marginal**

# PERMUTATION IMPORTANCE

RMSLE increase when each feature is shuffled — what the tuned model actually uses

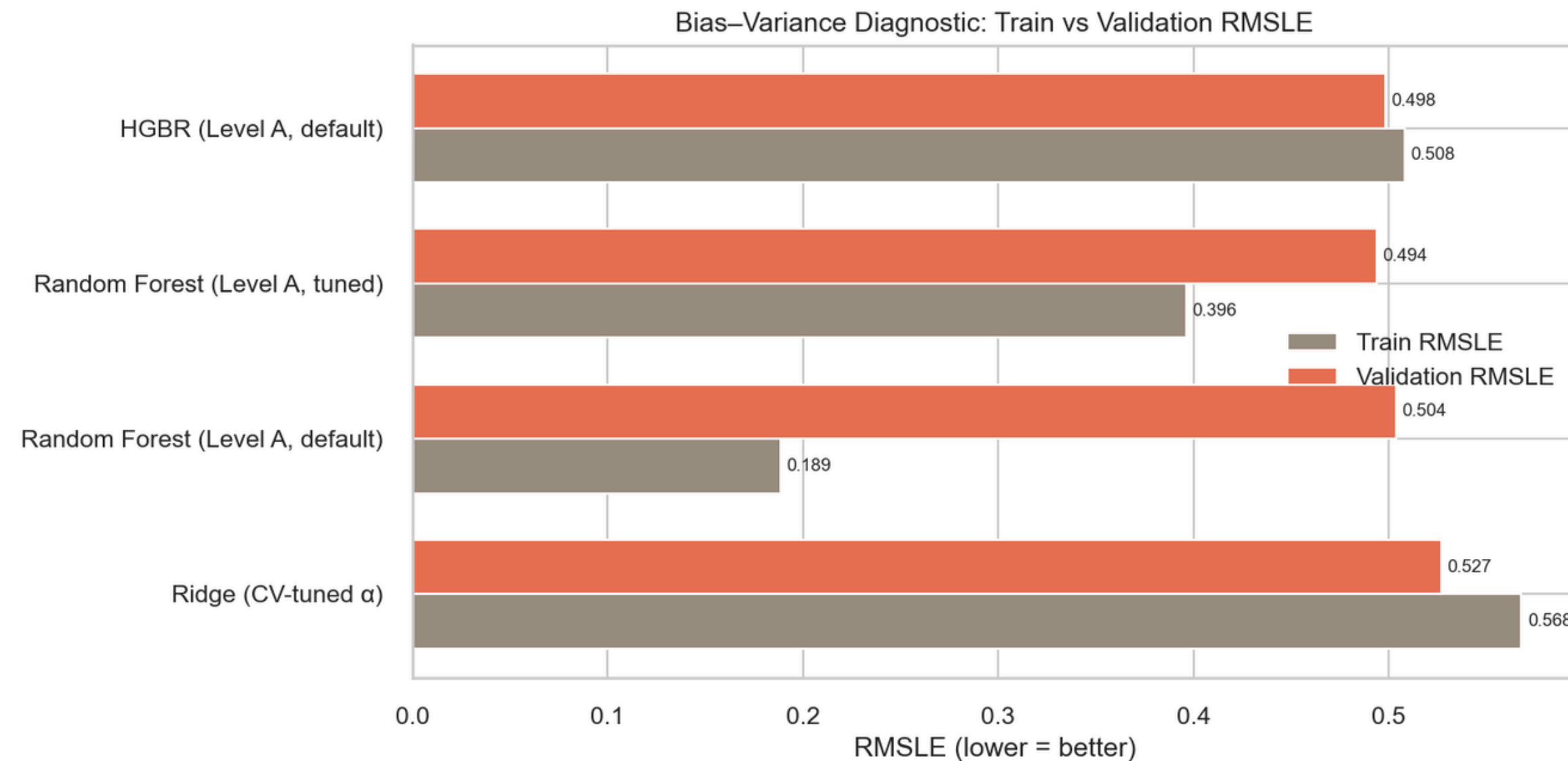


## WHAT DRIVES THE MODEL

- **Temporal features dominate** as rolling\_mean\_12 (0.143), lag\_4 (0.084), lag\_8 (0.049) — the top three carry most predictive power.
- **rolling\_mean\_12 is ~1.7× the next** meaning every non-temporal predictor scores below 0.02.
- **Promotion & price are secondary** as discount\_pct, emailer, homepage — they modulate demand around the temporal baseline.

# BIAS-VARIANCE DIAGNOSTICS

Train vs validation RMSLE — where each model sits on the spectrum



## READING THE GAPS

- **RF default** — **gap +167%** means severe overfitting (unbounded depth, leaf = 1).
- **RF tuned** — **gap +25%** means variance controlled; accuracy barely moves (0.498→0.508).
- **HGBR default** — **gap -2%** means near-ideal generalisation out of the box.
- **Ridge** — **negative gap** means controlled underfitting; the additive ceiling.

*Tuning's job here is variance reduction, not accuracy — a third signal of the predictability ceiling.*

# CLUSTER-SEGMENTED MODELLING (LEVEL B)

One HGBR per cluster vs one global HGBR — hyperparameters fixed to isolate segmentation

LEVEL A — GLOBAL

**0.4984**

[0.4926, 0.5044]

LEVEL B — SEGMENTED

**0.4985**

[0.4925, 0.5045]

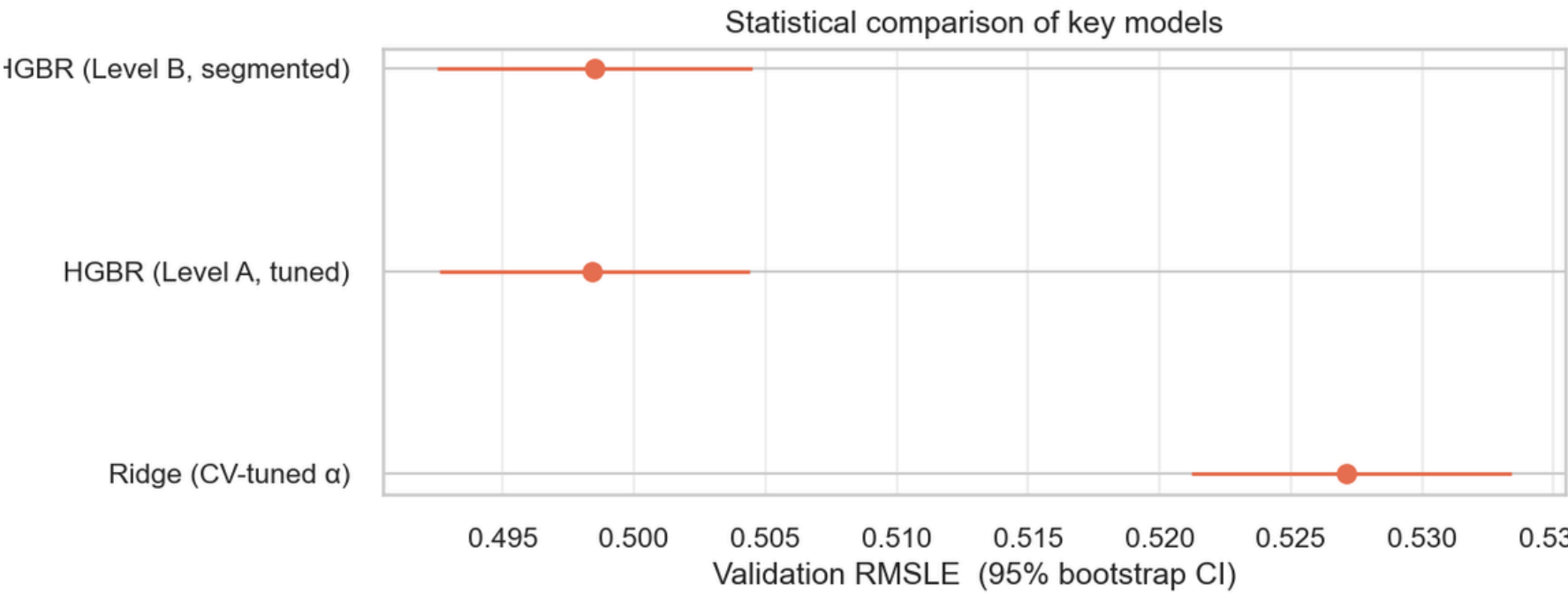
DIFFERENCE

**-0.0001**

CIs overlap → not significant

PER-CLUSTER Δ (Level B – Level A)

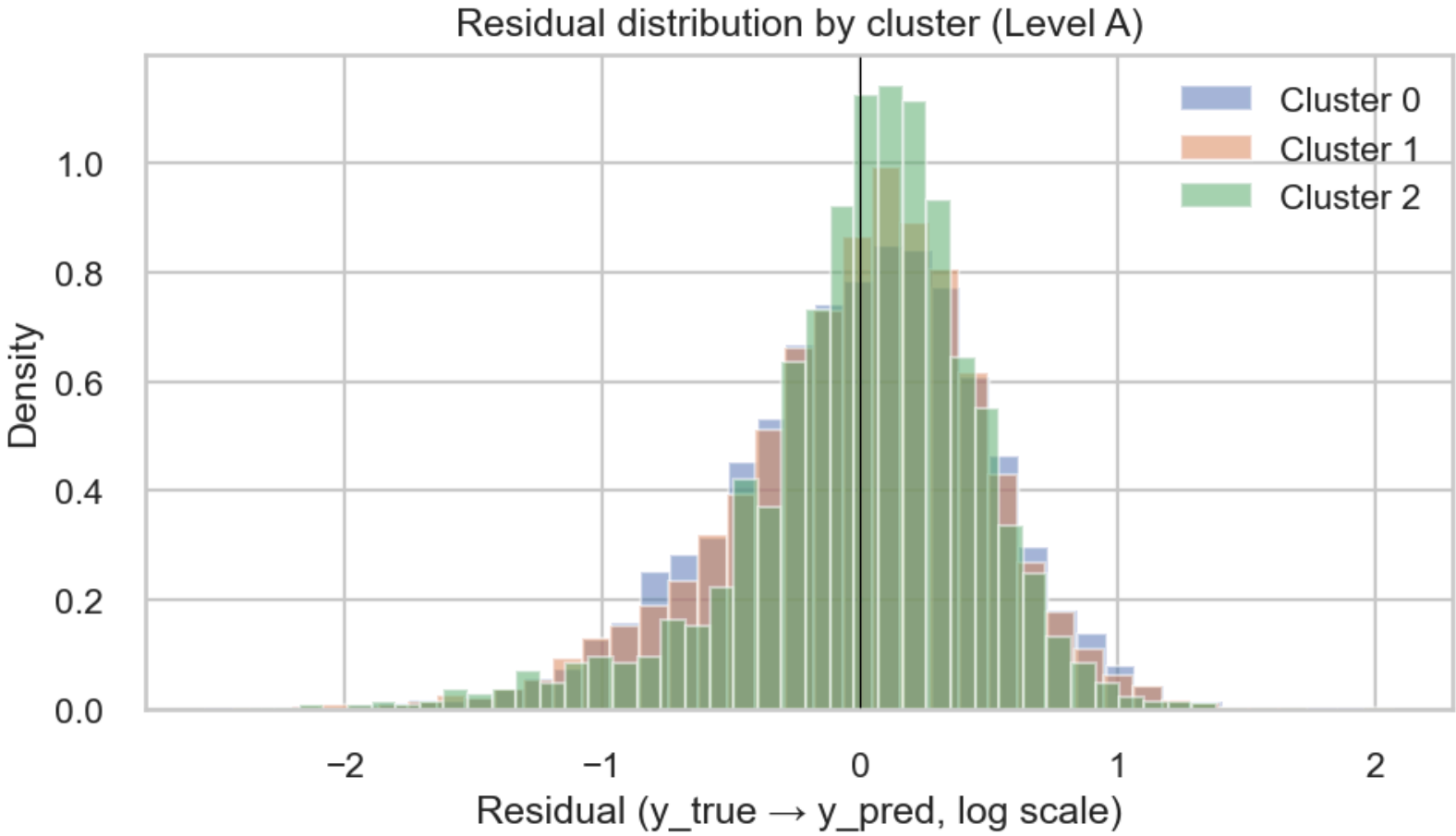
| CLUSTER    | PROFILE                        | Δ RMSLE |
|------------|--------------------------------|---------|
| 0 · Type B | high volume, low volatility    | -0.0006 |
| 1 · Type C | low volume, highest volatility | +0.0006 |
| 2 · Type A | mid volume, price-sensitive    | -0.0007 |



The hypothesised partial-pooling bias does not appear. **The global HGBR already holds cluster\_id and the temporal features (lag\_4/8, rolling\_mean\_12) that implicitly encode each center’s demand regime — so physical segmentation adds essentially nothing.**

# DIAGNOSTICS: WHY CLUSTER 1 IS HARDEST

Residual analysis: is the hardest cluster biased, or just intrinsically noisy?



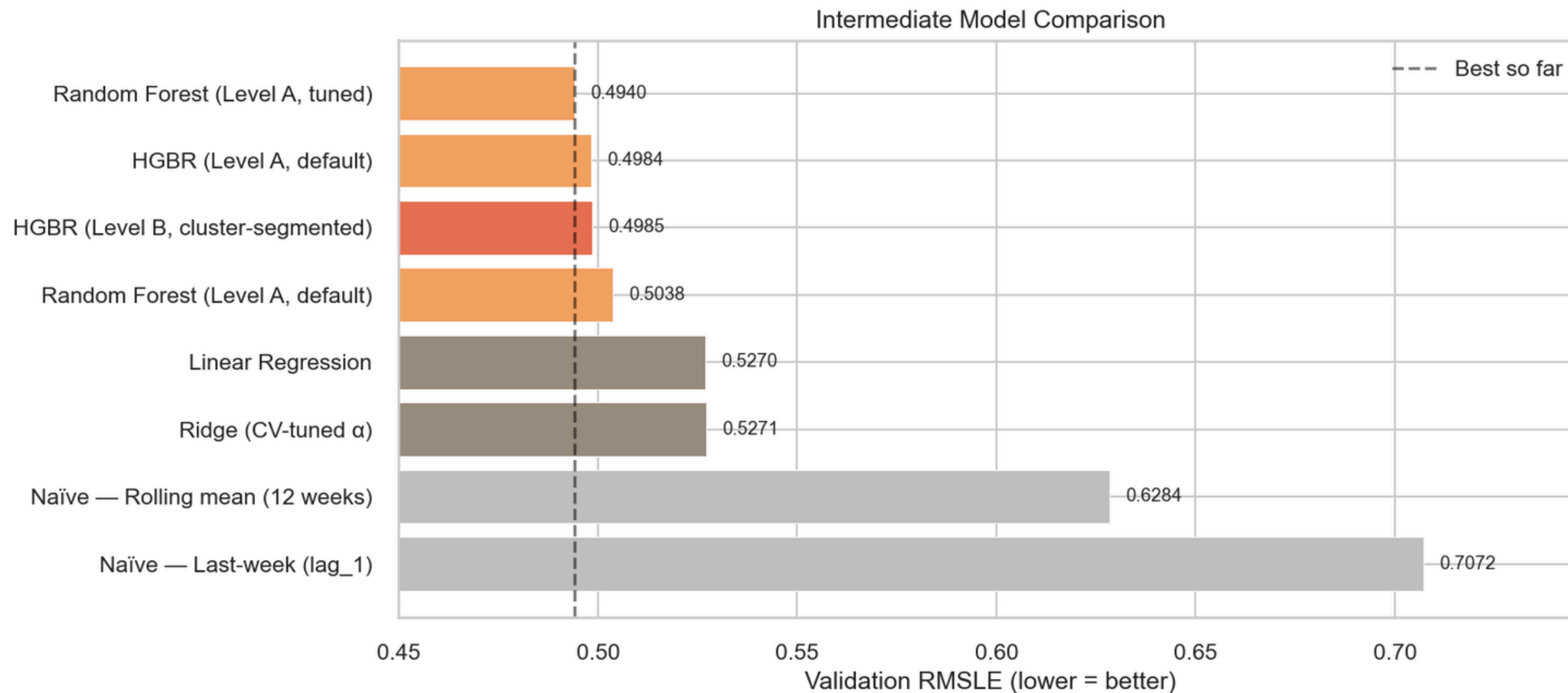
| Cluster                                                                                        | Mean | CV   | RMSLE |
|------------------------------------------------------------------------------------------------|------|------|-------|
| 0                                                                                              | 291  | 1.22 | 0.461 |
| 2                                                                                              | 222  | 1.18 | 0.499 |
| 1  noisiest | 179  | 1.12 | 0.518 |

## IT'S A PREDICTABILITY CEILING, NOT A GAP

- **Unbiased everywhere** — mean residuals  $\approx 0$  in all clusters (0.018, -0.020, -0.021).
- **Cluster 1 = highest noise-to-signal** — largest fluctuations relative to its low demand — exactly what RMSLE penalises.
- **No structure in residuals** — fan-shaped scatter, not systematic over/under-prediction — little signal left to exploit.

# INTERMEDIATE SUMMARY

Three layers of separation summarise the whole pipeline so far



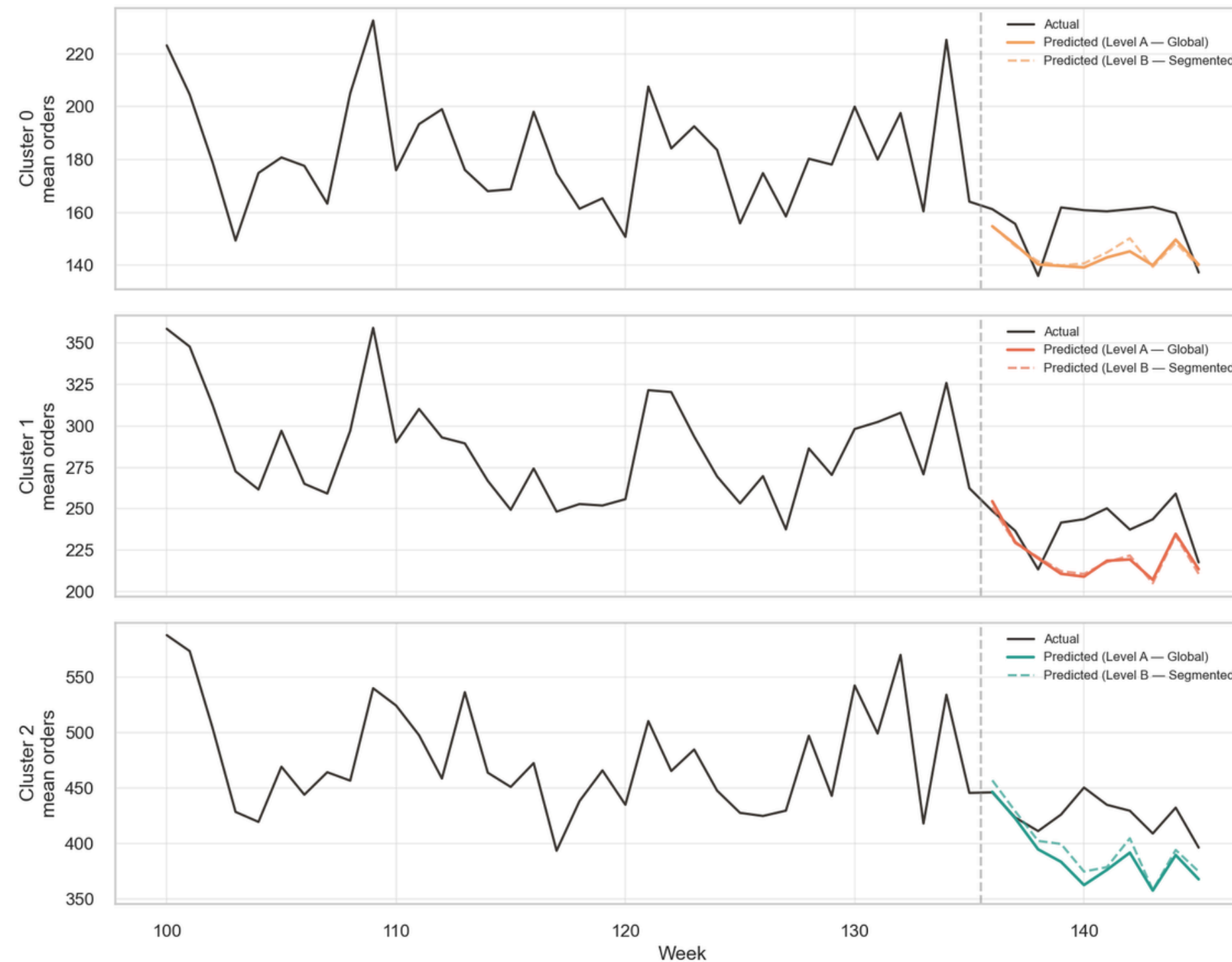
## KEY OBSERVATIONS

- **Naïve floor (0.7072 / 0.6284)** — every learned model clears it comfortably.
- **Linear plateau (~0.527)** LR, Ridge, cluster-Ridge all cluster here — the additive ceiling.
- **Non-linear plateau (~0.494–0.508)** all 4 ensemble configs within 0.0125 RMSLE — not statistically separable.
- **Hierarchy: Naïve < Linear < Non-linear** best model is 20.9% better than the floor; segmentation effect stays tiny.

# ACTUAL vs PREDICTED BY CLUSTER

Level A (solid) vs Level B (dashed) over the validation window, weeks 136–145

B.5 Actual vs Predicted Demand — Level A vs Level B, by Cluster



## WHAT THE CURVES SHOW

- **A and B are indistinguishable** - solid and dashed lines overlap so closely they read as one trajectory in every cluster.
- **Level & volatility captured** - models track each archetype's typical demand band well.
- **Shocks are not anticipated** - week-to-week deviations remain — the residual is irreducible uncertainty, not missing complexity.
- **Confirms the null result** - cluster-aware modelling does not change the forecast under the full feature regime.

# CLUSTER-RIDGE vs GLOBAL HGBR

Asymmetric Format 4 test: can structural specialisation replace model capacity?

3× Cluster-Ridge

0.5262

~7% worse · bootstrap CIs do NOT overlap

Global HGBR

0.4984

PER-CLUSTER Δ (HGBR → Ridge)

| Cluster | Ridge  | HGBR   | Δ      |
|---------|--------|--------|--------|
| 0       | 0.5356 | 0.5119 | -0.024 |
| 1       | 0.5290 | 0.4963 | -0.033 |
| 2       | 0.4841 | 0.4577 | -0.026 |

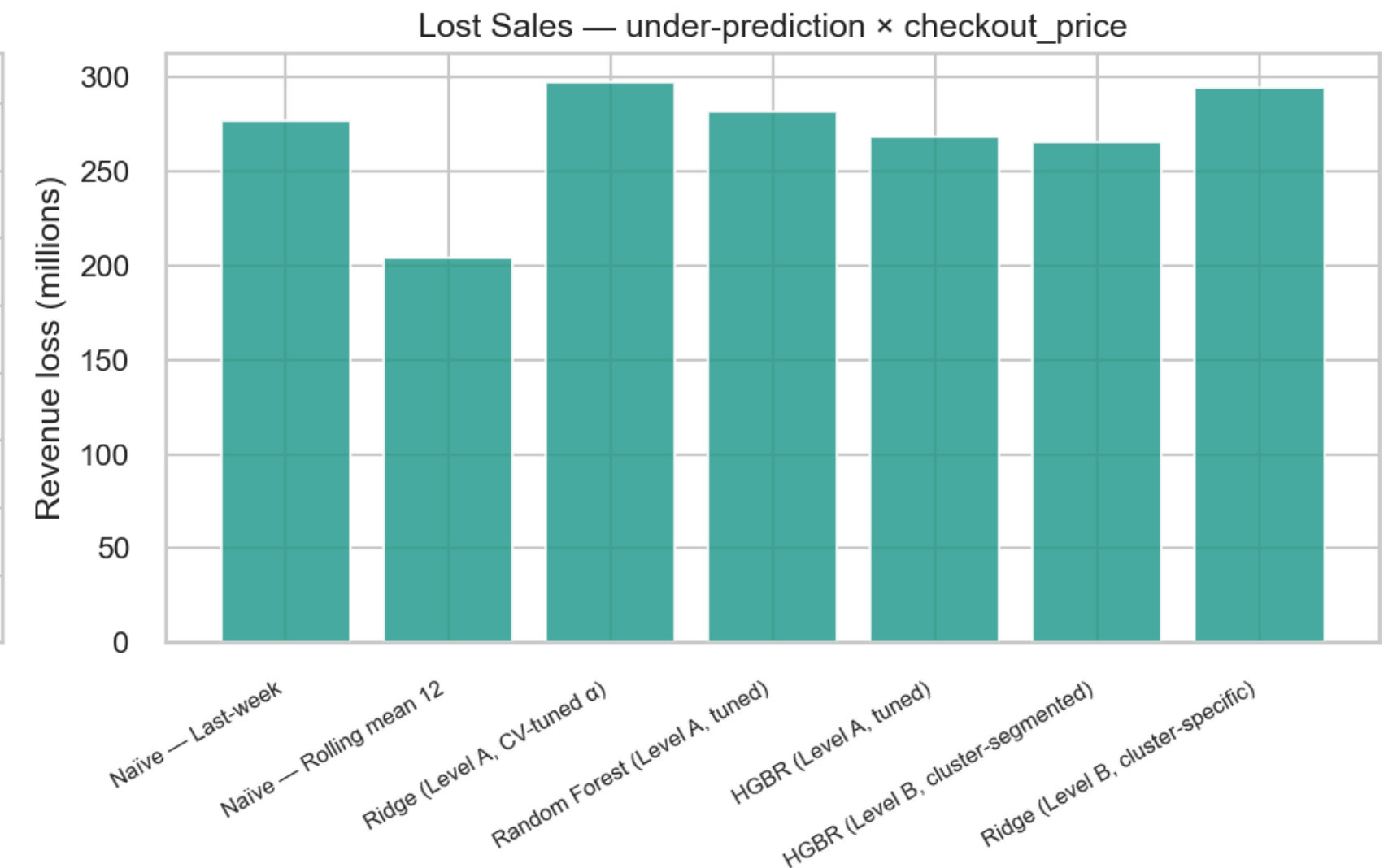
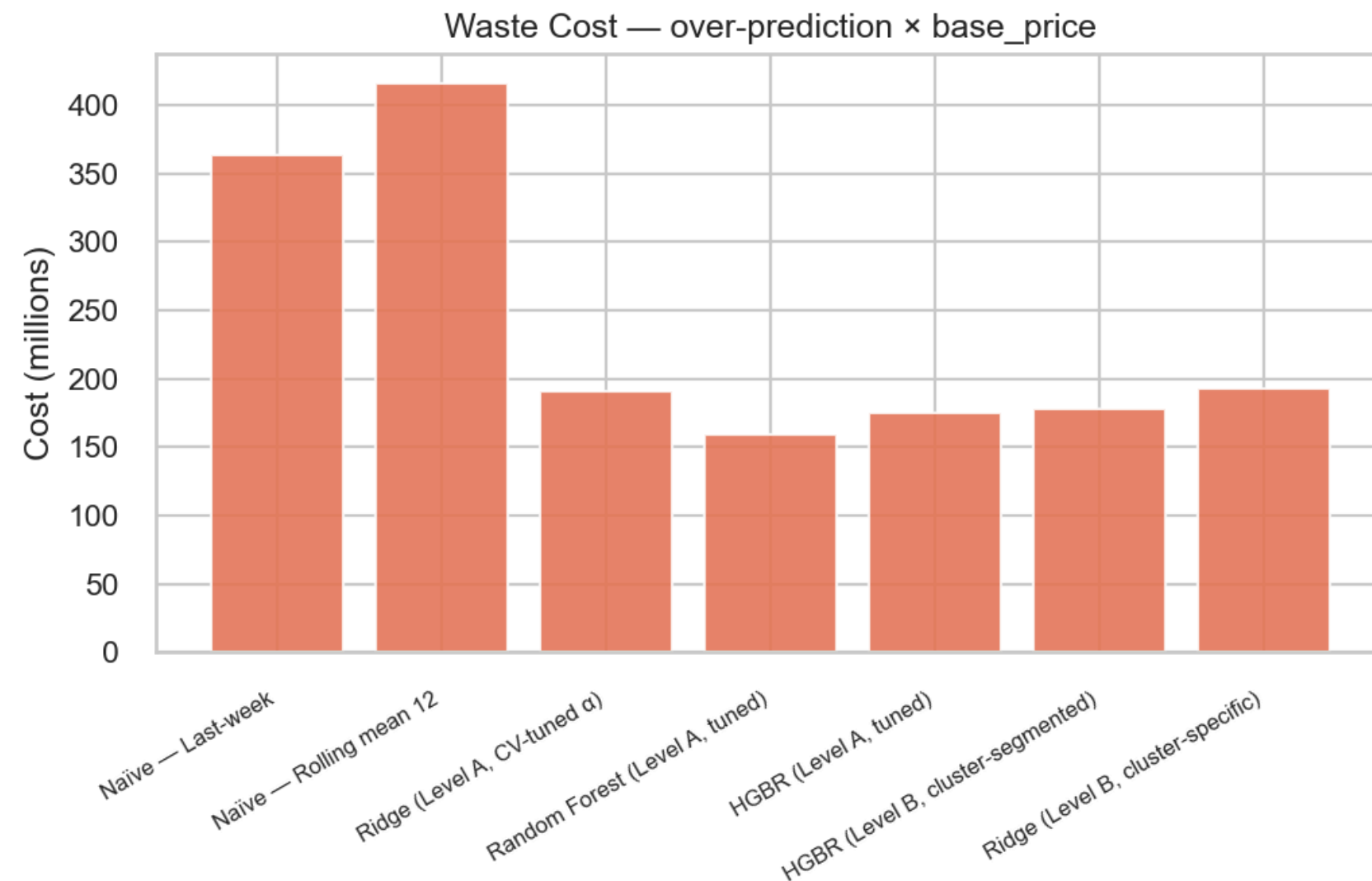
## SPECIALISATION ≠ CAPACITY

- **Ridge loses in every cluster** - the degradation is broad, not confined to one problematic segment.
- **The win is non-linear interactions** - price × promotion × temporal structure that no additive model can represent, however finely segmented.
- **Confirms the through-line** -  $A \approx B$ : gap stable without the dominant feature and Ridge < HGBR.
- **Model class is the lever** - the decisive jump is linear → non-linear, not partitioning the data.

# BUSINESS IMPACT LAYER

Translating forecast error into euros — over-prediction → waste, under-prediction → lost sales

Business Impact by Model — Validation Set

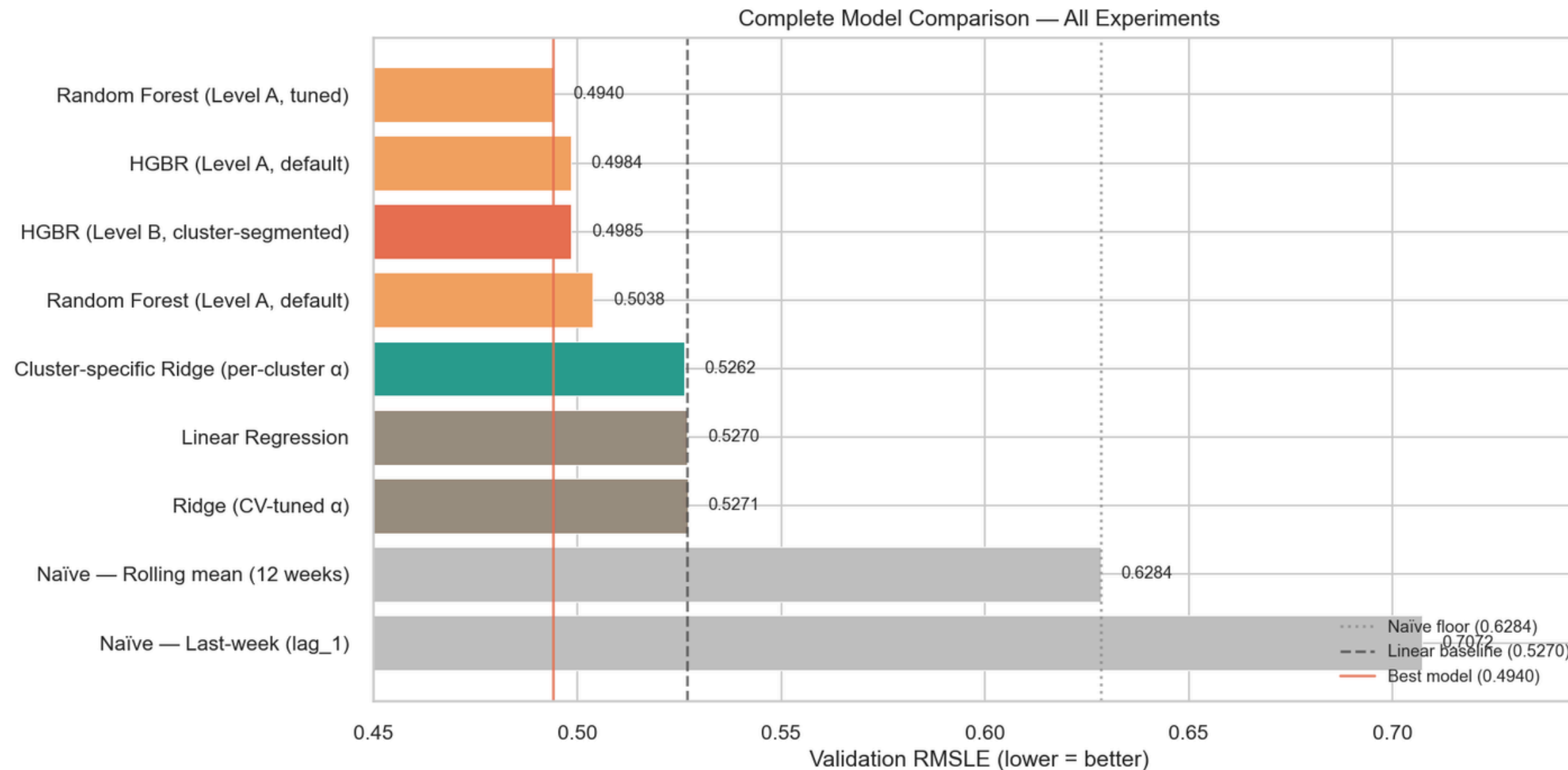


## COST SAVINGS

- **Level B costs ~1.8% more waste**  
segmentation gives no operational edge either.
- **HGBR A / B → 30.7%**  
all learned models collapse into one cost band.
- **Random Forest A → 31.1% best**  
operational result vs the worst naïve baseline (€640M).
- **Consistent shortage bias** - all good models under-predict slightly — lost sales exceed waste.

# FINAL SUMMARY — ALL EXPERIMENTS

The complete model ladder, from naïve floor to tuned ensembles



## THE ANSWER

- **RQ verdict: not distinguishable** as global 0.4955 vs segmented 0.4971 — inside the bootstrap noise band.
- **Pipeline justified: –20.9%** as linear adds ~15%, non-linear a further ~7% — each layer earns its place.
- **Model class is the dominant lever** as linear→non-linear is the only significant jump; segmentation never leaves the plateau.
- **Format 4 confirms it** cluster-Ridge loses to global HGBR by ~7% with non-overlapping CIs.

## CONCLUSION

**Segmentation does not reduce error —  
model class does.**

### PIPELINE JUSTIFIED

Best model cuts RMSLE 20.9% vs the naïve floor — gains accumulate at every layer.

### FORMAT 4 CONFIRMS

Cluster-Ridge loses to global HGBR by ~7% (non-overlapping CIs). Specialisation ≠ capacity.

### MODEL CLASS WINS

Linear → non-linear is the only significant jump. The 4 non-linear configs sit within 0.0125 RMSLE.

### WHERE IT HELPS

Only Cluster 1 (highest volatility) gains from a local model once the dominant feature is removed.

*Contribution: clustering is the wrong lever for accuracy on this dataset — it earns its place as an interpretability and diagnostics layer.*



# Thank you

*Teodora Atanasova & Martina Puglisi*